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Visualize CSV Data with Node-RED on ctrlX CORE

Sepp-31

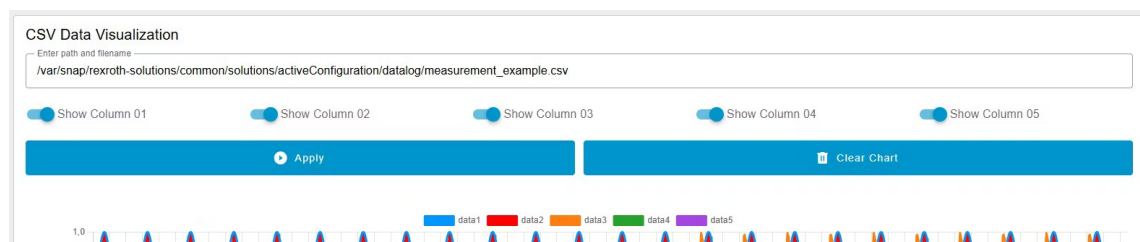
55–69 minutes

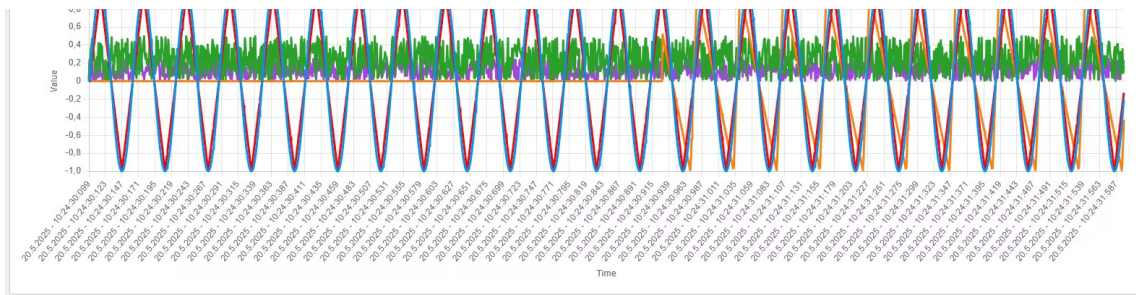
30/10/2025

How-To | HMI | Data Visualization

Overview

This How-To will show you, how you can use Node-RED to visualize data from a CSV file, that is stored on the ctrlX CORE. The CSV file may have been created by your PLC program or another app, but for test purposes you can simply download the same CSV file I used at the end of the article and upload it to your ctrlX CORE. Also the entire Node-RED flow can be downloaded at the end of the article. It contains the configuration for a dashboard, that looks like the image below. On the dashboard you can specify the path to your CSV file and select which columns of the file should be visualized. In this example, the dashboard is pre-configured to handle five data columns.





Node-RED Dashboard 2.0

Versions used

ctrlX CORE Runtime 3.6.0

Node-RED v4.0.9-ctrlx

Dashboard 2.0

Format of CSV File

The CSV file contains 6 columns. The first column represents the time scale in Unix time. The other 5 columns contain data from measurements of five signals recorded over a period of 1500 milliseconds. You can download the example file at the end of the article (in the ZIP folder).

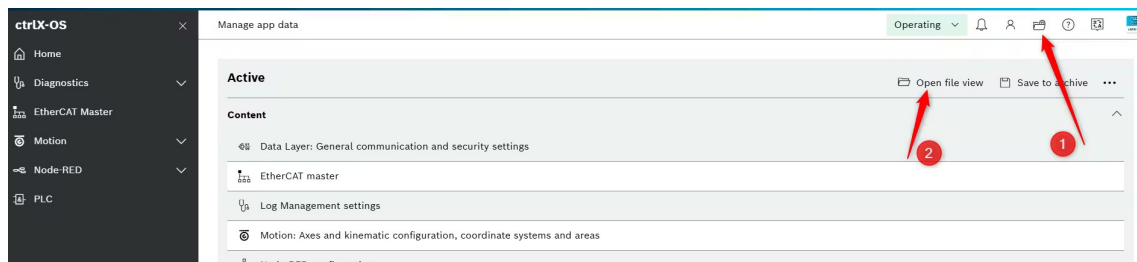
	A	B	C	D	E	F
1	time	data1	data2	data3	data4	data5
2	1747729470099	0	0	0	0	0
3	1747729470100	0	0	0	0.1922733	0.2370247
4	1747729470101	0.10036171	0.06400001	0	0.2501909	0.11312276
5	1747729470102	0.1997531	0.128028	0	0.3694766	0.1523368
6	1747729470103	0.2970596	0.192012	0	0.4144137	0.05975245
7	1747729470104	0.3913332	0.255972	0	0.477006	0.1845052
8	1747729470105	0.4816326	0.319912	0	0.24204	0.01000813
9	1747729470106	0.5670775	0.383852	0	0.3360247	0.10225829
10	1747729470107	0.6468739	0.447848	0	0.2328788	0.00653367
11	1747729470108	0.7201172	0.511824	0	0.10465705	0.08947465

Content of CSV file

Storage location

In this example, the CSV file is stored on the internal storage of a ctrlX CORE. Go to "Manage App Data" (1) and click "Open

File View" (2).



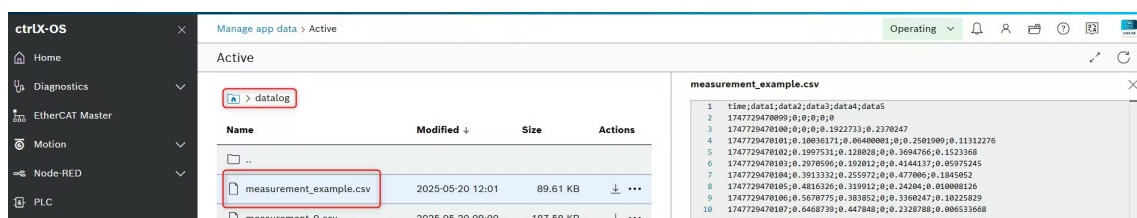
Manage app data

There you can create a new folder by clicking on the "+" button (3) in the top right corner and name it "datalog". Then open the folder (4).



Create new folder

There you can upload the example CSV file with the arrow button in the upper right corner and should then be able to preview the file.

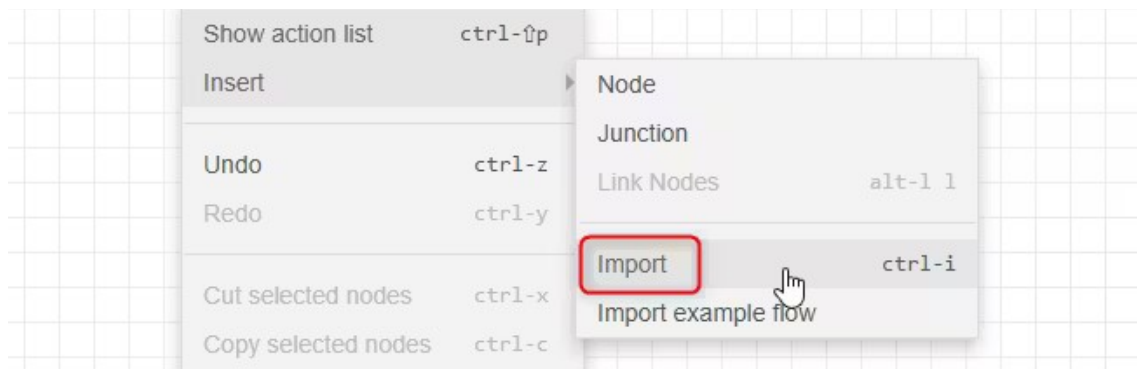


File preview

Node-RED flow

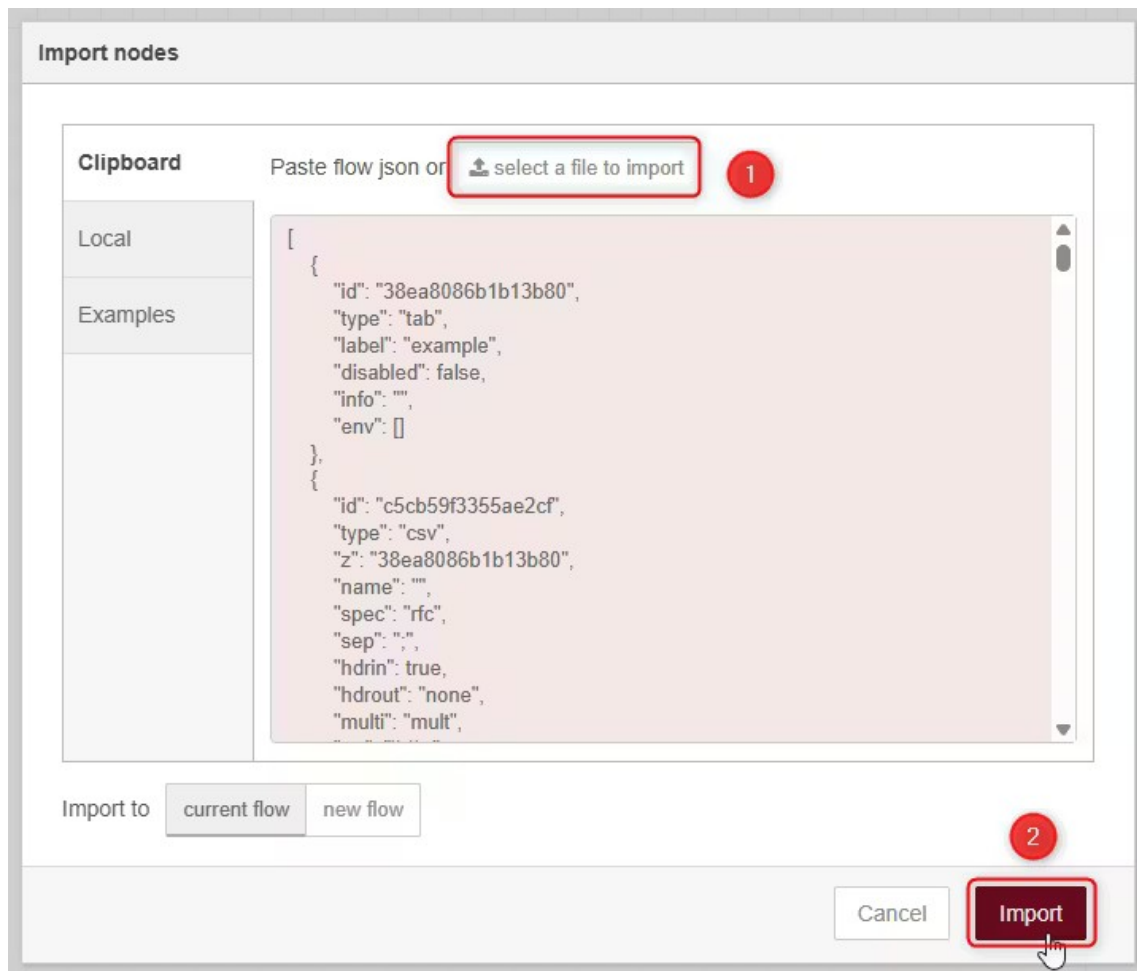
First download the file "CSV_Data_Visualization.json" attached in the end of the article (in the ZIP folder). Then open the Node-RED flow editor. There right click in the empty editor and go to "Insert" and click "Import".





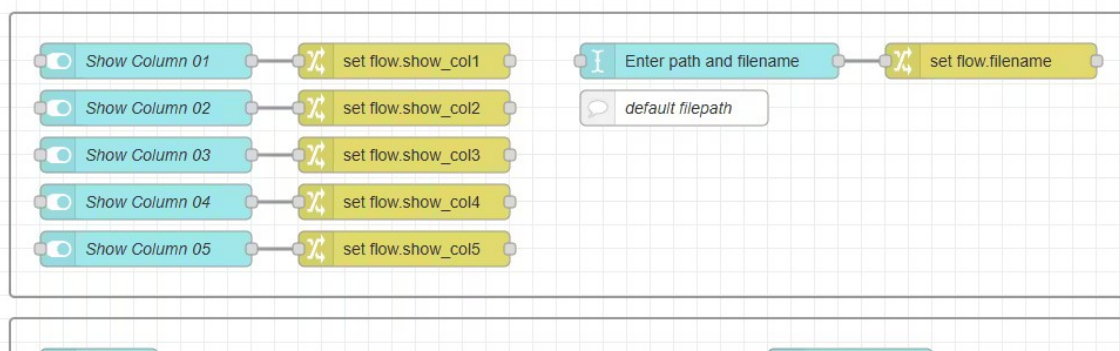
Import Node-RED flow

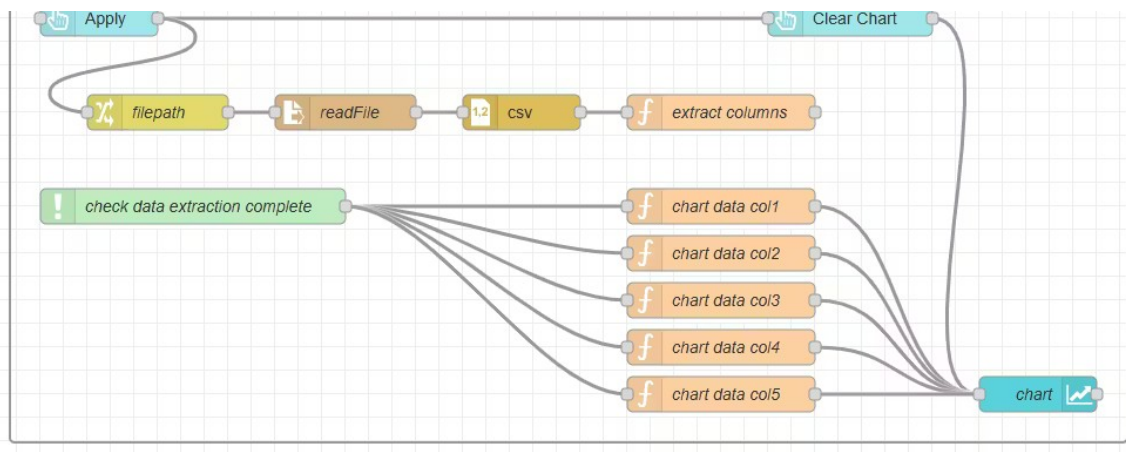
Select the file (1) you downloaded and click "Import" (2).



Import mask

Now the following nodes should appear in your flow.





Node-RED flow

Let me explain, what these nodes do. In the upper section of the flow, you see five switch nodes, that are used for column selection, which data columns should be displayed on the dashboard. The boolean values are then stored as flow variables, so that the selections can be used later in the code. On the right you see a text input node, where you can input the path to your CSV file, also then saved to a flow variable. Note that the path to the CSV file, that you uploaded to the "datalog" folder earlier will be `"/var/snap/rexroth-solutions/common/solutions/activeConfiguration/datalog/measurement_example.csv"`.

In the lower section of the code you first see the two buttons "Apply" and "Clear Chart". When "Clear Chart" will be clicked on the dashboard an empty `msg.payload` will be send to the chart node, which will then clear it. When "Apply" will be clicked the following four nodes will be triggered:

- "filepath" reads the file path from the flow variable and sends it as `msg.payload` to
- "readFile", where the file will be read and outputted as a long string to the
- "csv" node, where the file is interpreted: the column names will be read from the first row of the file and the time and data

columns will be extracted.

- "extract columns" is then a function node, where JavaScript Code will store the names of the data columns as an array in the flow context and also the time column and the five data columns as arrays in the flow context

Once this is completed the green node you see below will trigger the five function nodes that will send pairs of time and the corresponding data to the chart, depending on which data columns are selected to be displayed on the dashboard. The pairs of time and data value are passed via `msg.payload` and the name of the data series is passed via `msg.topic`. So it will look like this:

```
msg.payload = [ {"time":1747729470100,"value":0},  
  
{"time":1747729470101,"value":0.10036171},  
  
{"time":1747729470102,"value":0.19975310},  
                //...  
                ];  
msg.topic    = "data1";
```

Finally, a chart node (ui-chart) displays the selected data curves. The chart node is configured in append mode, with the x-axis set to timescale. Ensure that the x-axis limit is set sufficiently high to display all data points from the CSV file. In the "Series" properties, `msg.topic` is used, and the X and Y value fields are configured to use "time" and "value", respectively, as sent in `msg.payload` from the preceding function nodes.



Type

Line

Show Legend

☒

Interpolation

Linear

Action

Append

Point Style

None

Radius (px)

4

Axis Configuration

X-Axis Type

Timescale

Format

▼

A

D - HH:mm:ss:u

X-Axis Label

Time

X-Axis Limit

last

500

Weeks

▼

OR

1000

points

Y-Axis Label

Value

Y-Axis

min.

max.

Properties

Series:

▼

msg.topic

X:

▼

key: time

Y:

▼

key: value

Chart node

CSV_Visualization.zip

36.42KB

CAM CURVE

1 of 1

G

Types

How-to

Products

HMI

Markets

Manufacturing

Assembly Lines

Building Automation

Logistic

Packaging

Printing

Production Machines

Robotics

Semicon & Electronics

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👉 Do you know a device that is missing here? You're welcome to share it with the community – please fill out directly our form here. ctrlX World Partner can enter their devices directly via the ctrlX OS Ecosystem Portal. Vendor overview Interoperability status LIMITED - only limited usage possible OK - no known problems NO - not functioning with not functioning with ctrlX OS EtherCAT Master [table] Vendor Product Description Communication App Status How-to Arduino Arduino Nicla Sense ME low-power development board equipped with advanced sensors for motion and environmental monitoring ctrlX Standard USB OK How-to Balluff Balluff BIS00U9 EtherCAT Multi-frequency RFID processors ctrlX EtherCAT Master OK How-to Balluff BIS0147 EtherCAT Multi-frequency